

CLINOMA PLATFORM

MATCHING ASSESSMENT

Day 2 Pediatrics Quiz Booklet

Interactive Diagnosis & Classification

Prepared by: Clinoma Platform Team

Set 1: Childhood Obesity (Differential Diagnosis)

Column I (Questions)	[Match]	Column II (Answers)
1. Nutritional Obesity	[]	A. Decreased/decelerated linear growth. Test for thyroid hormones.
2. Endocrine Obesity	[]	B. Consistent/accelerated growth, early puberty, advanced bone age (> 2 SD).
3. Genetic Syndrome Obesity	[]	C. Severe obesity < 5 years. Developmental delay, short stature, dysmorphic facies, hyperphagia.

Set 2: Congenital Hypothyroidism (Causes & Radiographic Features)

Column I (Questions)	[Match]	Column II (Answers)
1. Defect of fetal thyroid development (DYSGENESIS)	[]	A. Missing or too small gland.
2. Defect in thyroid hormone synthesis (DYSHORMONOGENESIS)	[]	B. Absent/small epiphyses, stippled appearance, cortical thickening.
3. Thyroid scan	[]	C. Defect in thyroid hormone synthesis.
4. Bone age (Greulich & Pyle method)	[]	D. Aplasia, Hypoplasia, Ectopia.

Set 3: Diabetes Mellitus (Insulin Therapy Management)

Column I (Questions)	[Match]	Column II (Answers)
1. Rapid/Short-acting (Lispro, Aspart)	[]	A. 30-60 mins before meals.
2. Regular	[]	B. 15 mins before meals.
3. Intermediate (NPH, Lente)	[]	C. 10-16 hours.
4. Long-acting (Glargine, Detemir)	[]	D. 20-24 hours.

Set 4: Short Stature (Normal Variants of Growth)

Column I (Questions)	[Match]	Column II (Answers)
1. Familial Short Stature	[]	A. Bone Age = Chronologic Age (Crucial differential point).
2. Constitutional Delay of Growth and Puberty (CDGP)	[]	B. Most achieve catch-up growth by 2 years of age to enter the normal range.
3. Idiopathic Short Stature (ISS)	[]	C. Normal birth size → downward shift at 3-6 months → grows parallel to but below the 3rd percentile by 3-4 years.
4. Small for Gestational Age (SGA) Infants	[]	D. Height $\leq$ 2 SD below the mean with no endocrine, metabolic, or systemic diagnosis.

Set 5: Hematological Hallmarks & Pathognomonic Features

Column I (Questions)	[Match]	Column II (Answers)
1. G6PD-deficient individuals	[]	A. Eosin-5-maleimide (EMA) is the diagnostic test of choice.
2. Hereditary Spherocytosis (HS)	[]	B. Hypocellular BM for age plus two of the following three criteria: Platelet count < 20,000/mm ³ , Absolute reticulocyte count < 40,000/mm ³ , or Absolute neutrophil count (ANC) < 500/mm ³ .
3. Hodgkin lymphoma (HL)	[]	C. The Reed-Sternberg (RS) cell is the hallmark diagnostic feature, characterized as a large cell (15-45 µm in diameter) containing multiple or multilobulated nuclei ("owl-eye" appearance).
4. Beta-Thalassemia Major	[]	D. Bone marrow aspiration and biopsy are confirmative and diagnostic if more than 25% of the bone marrow cells are lymphoblasts.
5. SEVERE APLASTIC ANEMIA CRITERIA	[]	E. Peripheral smear: TARGET CELLS, ANISOCYTOSIS, POIKILOCYTOSIS.
6. Acute Lymphoblastic Leukemia (ALL)	[]	F. Denatured hemoglobin aggregates are VISUALIZED as HEINZ BODIES in peripheral blood smears.

Set 6: Genetics & Inheritance Patterns

Column I (Questions)	[Match]	Column II (Answers)
1. Hemophilia A (Classic)	[]	A. MOST COMMON ENZYMATIC DISORDER of RBCs, inherited via SEX-LINKED RECESSIVE MODE on the X CHROMOSOME.
2. Hemophilia C	[]	B. Factor XI deficiency. (Autosomal recessive, rarest)
3. HYDROPS FETALIS	[]	C. Fanconi anemia and Diamond Blackfan syndrome.
4. Hereditary Spherocytosis (HS)	[]	D. Mode of inheritance: Autosomal Dominant (AD) in 75% of cases.
5. G6PD DEFICIENCY	[]	E. Factor VIII deficiency. (X-linked recessive, affects males)
6. CONGENITAL (Mostly Inherited) Etiology of Aplastic Anemia	[]	F. SEVERE HEMOLYSIS in utero, leading to DEATH (4 genes deleted).

Set 7: Clinical Manifestations & Unique Signs

Column I (Questions)	[Match]	Column II (Answers)
1. Iron deficiency anemia (IDA)	[]	A. Nonspecific signs including anorexia, fatigue, malaise, irritability, and intermittent low-grade fever.
2. IMMUNE THROMBOCYTOPENIA (ITP)	[]	B. SUDDEN APPEARANCE of PETECHIAL RASH, PURPURA, & ECCHYMOSES (BRUISING) in an otherwise healthy child.
3. BETA-THALASSEMIA MAJOR (COOLEY ANEMIA)	[]	C. pica, which is the intense craving for nonfood items like clay, dirt, rocks, starch, chalk, soap, paper, or cardboard.
4. Childhood Acute Lymphoblastic Leukemia (ALL) Initial presentation	[]	D. "THALASSEMIC FACIES" (frontal bossing, flat nasal bridge, maxilla hyperplasia)
5. Hemophilia in Mobile Children & Adolescents	[]	E. Hemarthrosis (Joint Bleeding): Most common site (up to 80% of hemorrhages). Affects ankles, knees, elbows.

Set 8: Laboratory Profiles & Investigations

Column I (Questions)	[Match]	Column II (Answers)
1. Iron profile in IDA	[]	A. INCREASED indirect bilirubin, high serum ferritin, high serum iron, high transferrin saturation.
2. Diagnosis in ITP	[]	B. Coagulation time & Activated partial thromboplastin time (aPTT): Prolonged
3. Blood chemistry in ALL	[]	C. CBC shows ISOLATED THROMBOCYTOPENIA
4. Biochemistry in Beta-Thalassemia Major	[]	D. shows markedly elevated lactate dehydrogenase (LDH) and uric acid.
5. Investigations in Hemophilia	[]	E. shows low serum iron, low serum ferritin, increased Total iron binding capacity (TIBC), and decreased transferrin saturation (TS).

Set 9: Severe Complications, Crises & Emergencies

Column I (Questions)	[Match]	Column II (Answers)
1. APLASTIC CRISIS	[]	A. IgG antibodies that neutralize clotting factor concentrates. Occurs in 30% of severe Hemophilia A.
2. MEGALOBlastic CRISIS	[]	B. MOST DANGEROUS! Almost universally triggered by Parvovirus B19. Completely halts marrow erythropoiesis.
3. HEMOLYTIC CRISIS	[]	C. Due to rapid, severe depletion of nutritional folate stores driven by chronic, highly accelerated bone marrow erythropoiesis.
4. Tumor Lysis Syndrome (TLS)	[]	D. Sudden, temporary acceleration of RBC destruction, precipitated by acute bacterial or viral infections.
5. Inhibitors (Major Complication)	[]	E. A life-threatening complication managed with aggressive IV hydration before chemotherapy, uric acid reduction via allopurinol, and close electrolyte monitoring.